

Advanced Shortest Paths: Bidirectional Dijkstra

Michael Levin

Higher School of Economics

Graph Algorithms
Data Structures and Algorithms

Outline

- 1 Bidirectional Search
- 2 Bidirectional Dijkstra

Shortest Path

Input: A graph G with *non-negative* edge weights, a source vertex s and a target vertex t .

Output: The shortest path between s and t in G .

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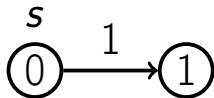
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- Millions of users of Google Maps want the result in a blink of an eye, all at the same time
- Need something significantly faster

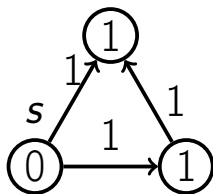
Dijkstra Progression

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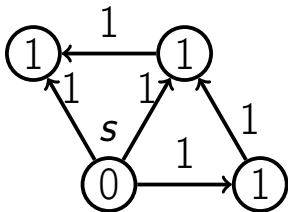
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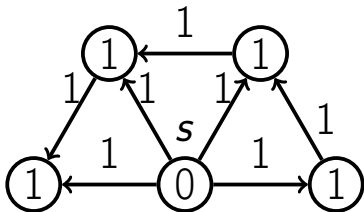
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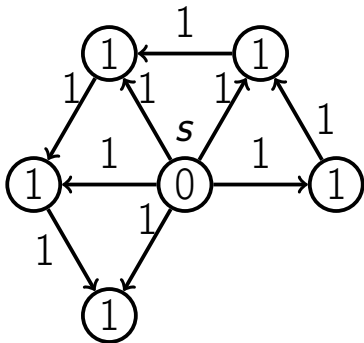
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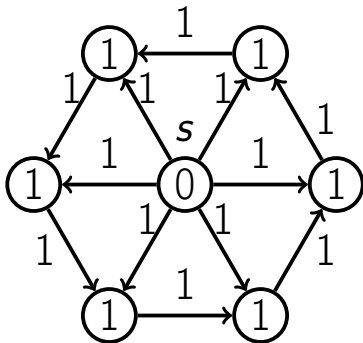
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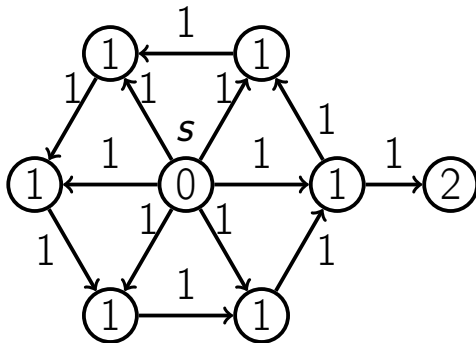
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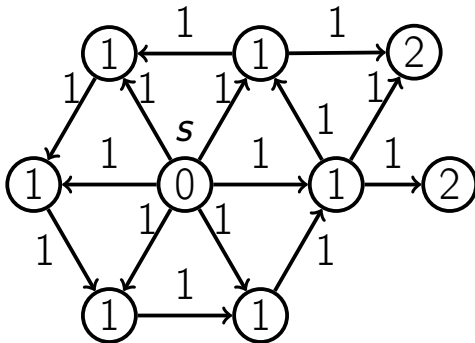
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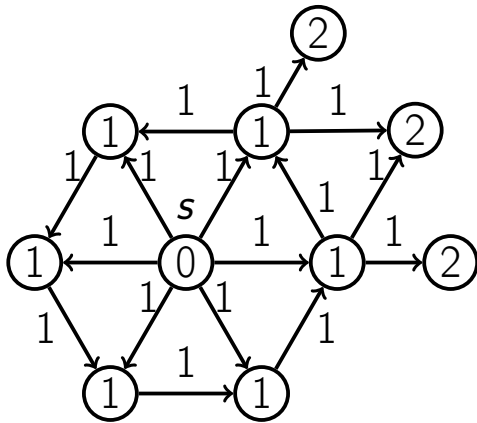
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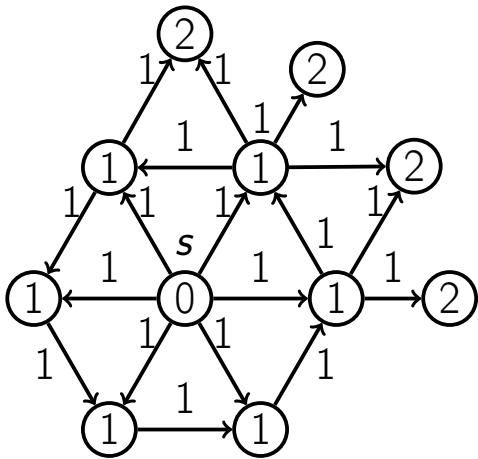
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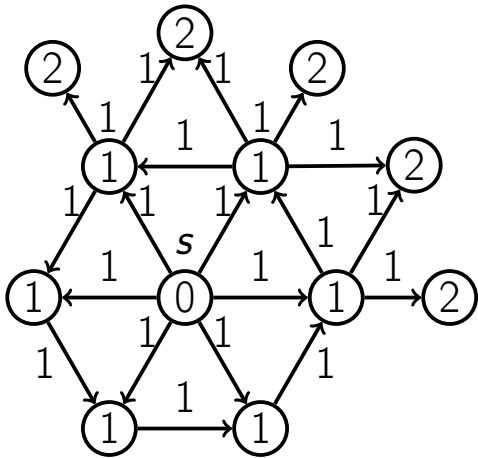
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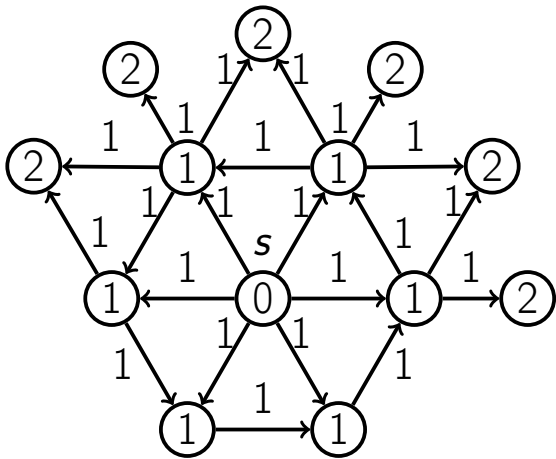
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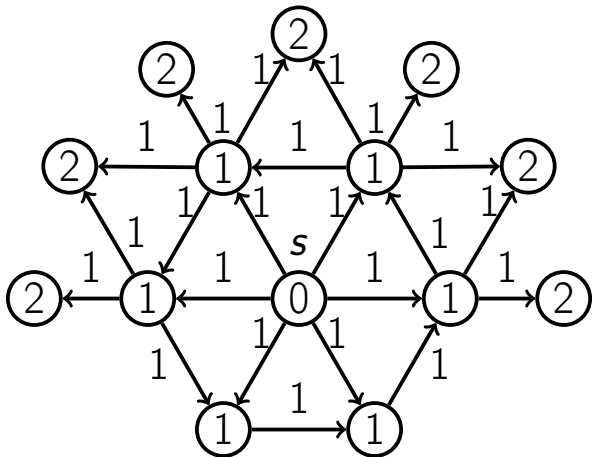
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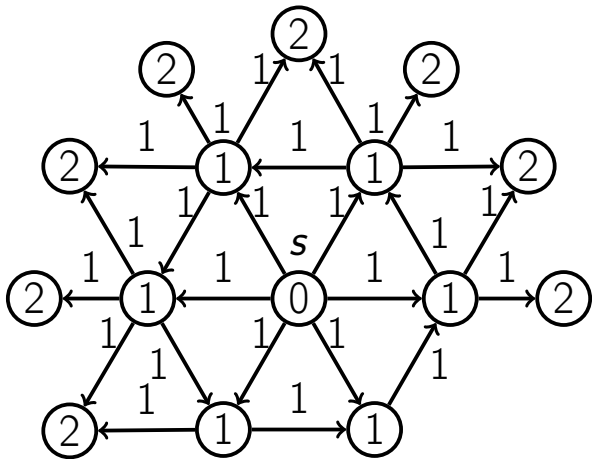
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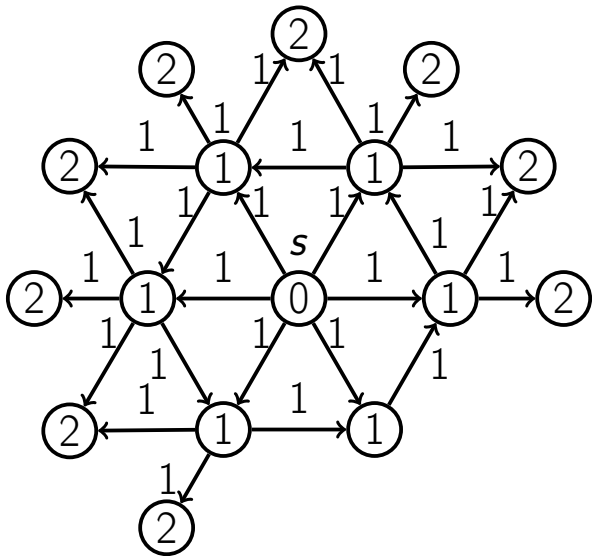
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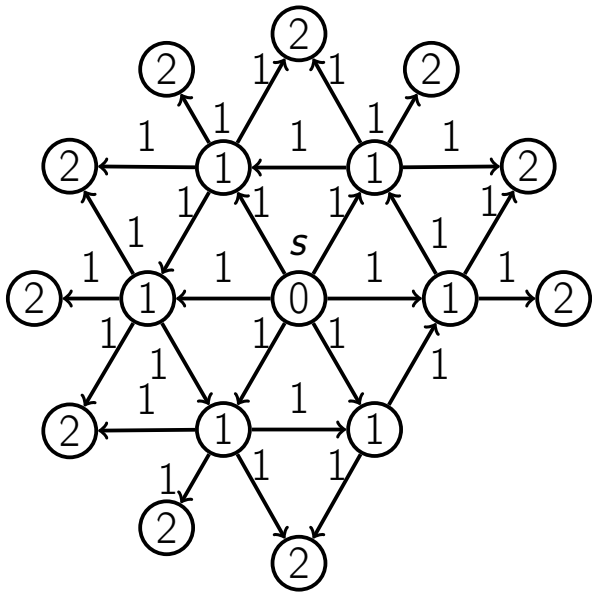
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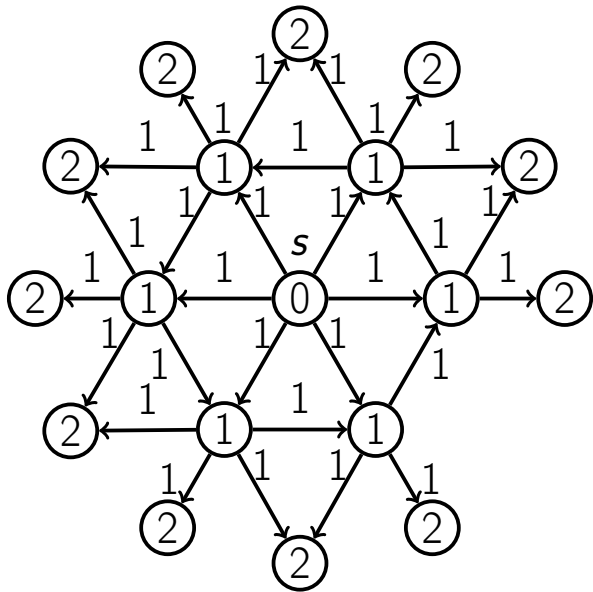
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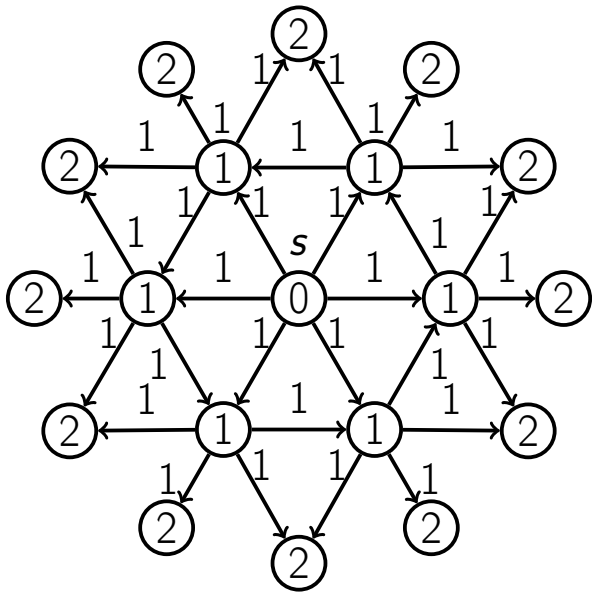
Dijkstra Progression



Dijkstra Progression



Dijkstra Progression



Idea: Growing Circle

Lemma

When a vertex u is selected via `ExtractMin`,
 $\text{dist}[u] = d(s, u)$.

- When a vertex is extracted from the priority queue for processing, all the vertices at smaller distances have already been processed

Idea: Growing Circle

Lemma

When a vertex u is selected via `ExtractMin`,
 $\text{dist}[u] = d(s, u)$.

- When a vertex is extracted from the priority queue for processing, all the vertices at smaller distances have already been processed
- A “circle” of processed vertices grows

Idea: Growing Circle

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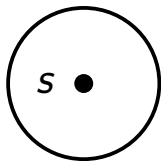
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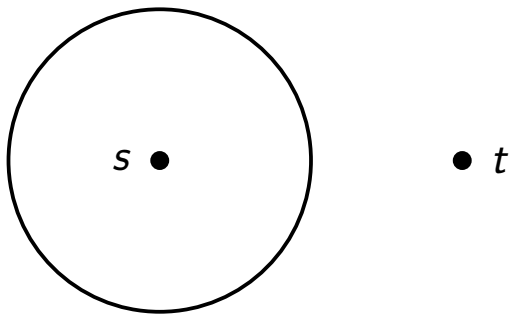
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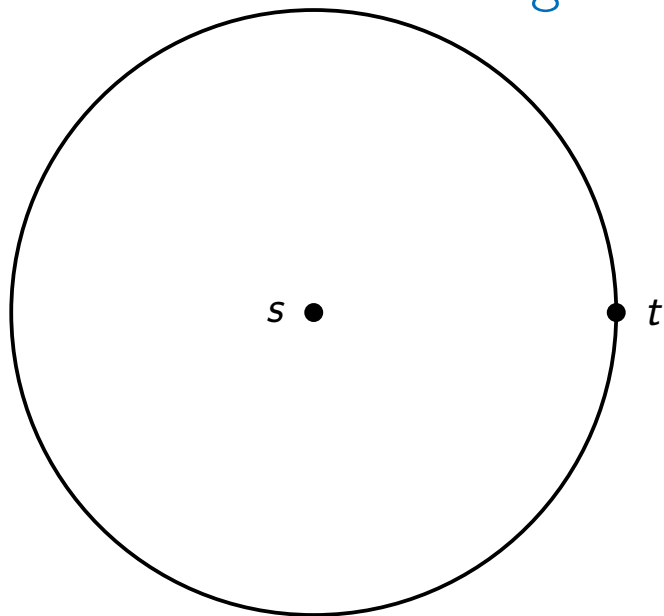
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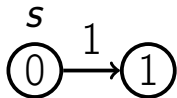


Idea: Bidirectional Search

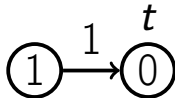
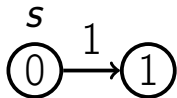
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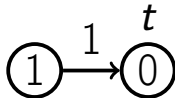
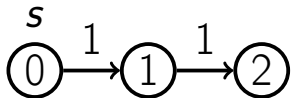
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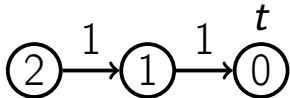
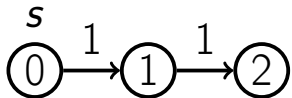
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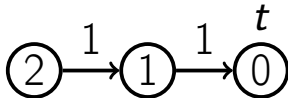
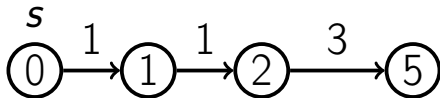
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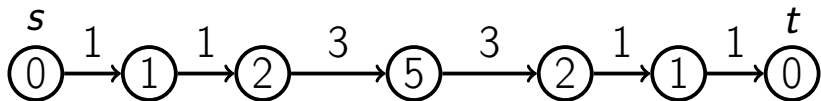
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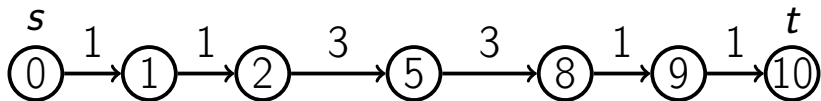
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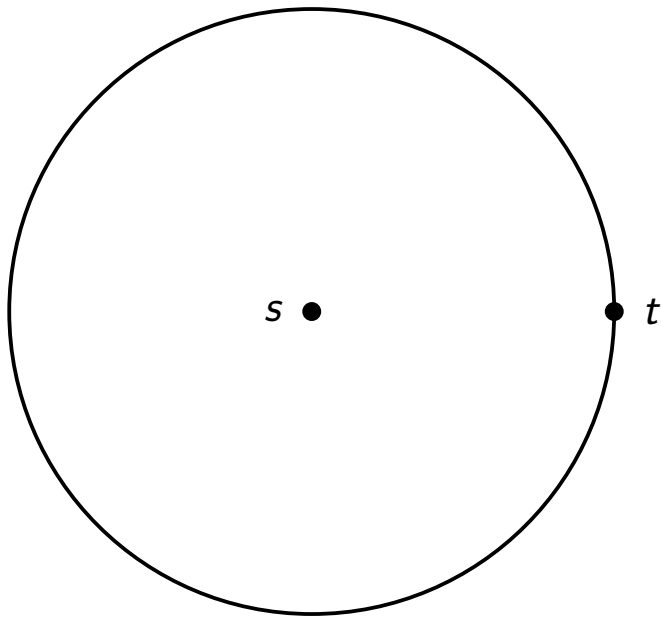
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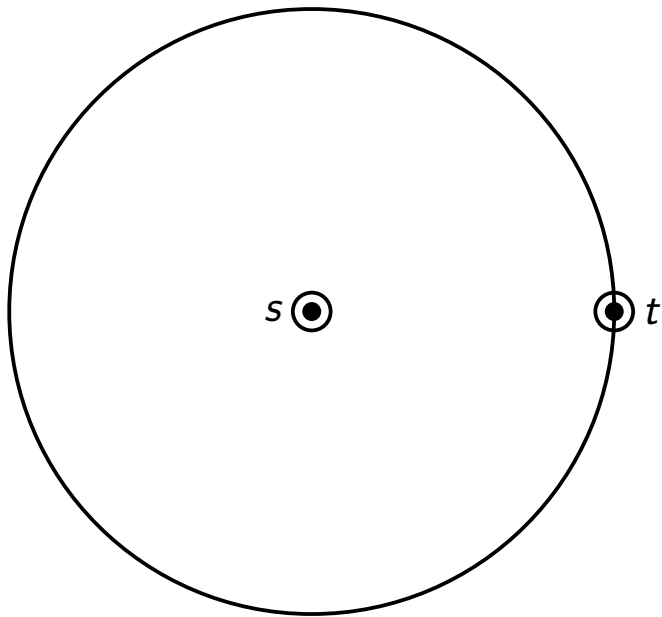
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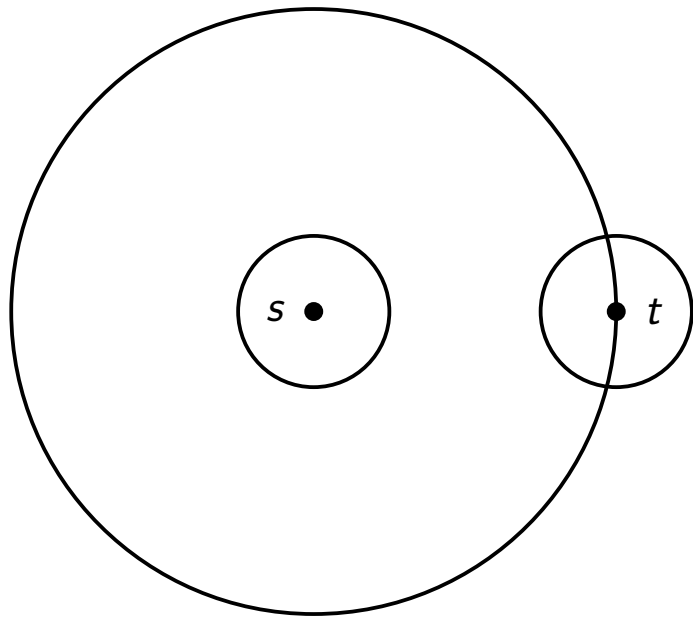
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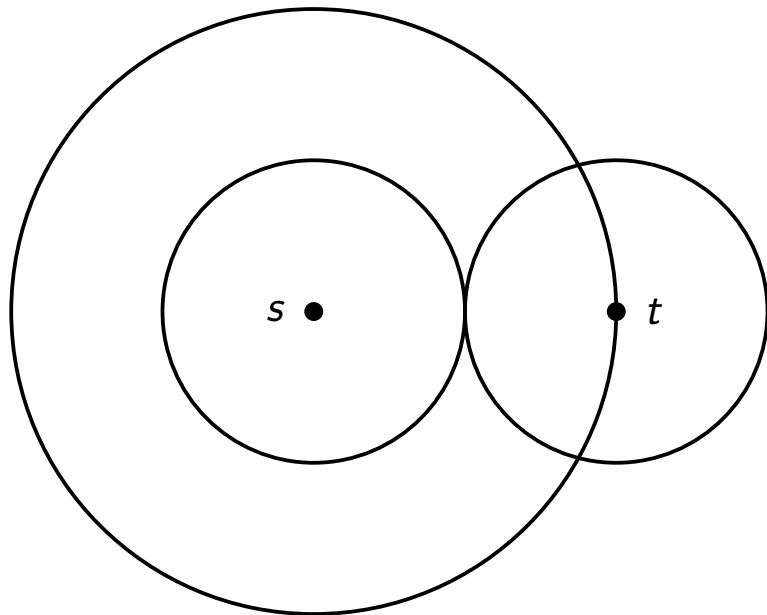
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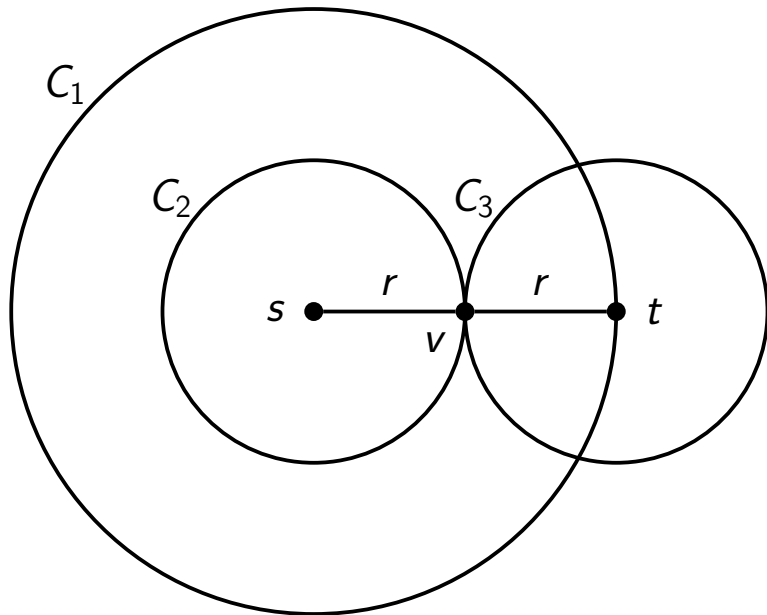
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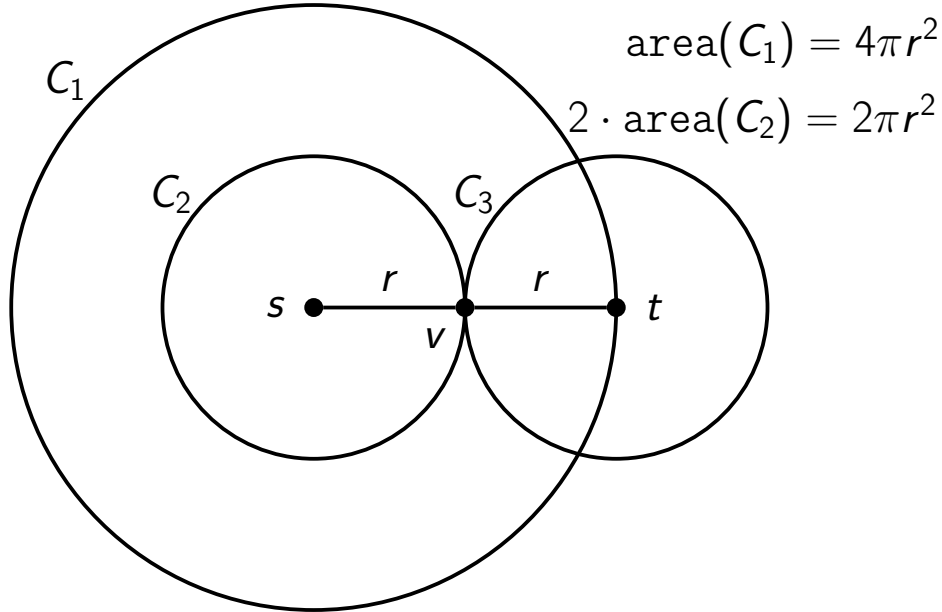
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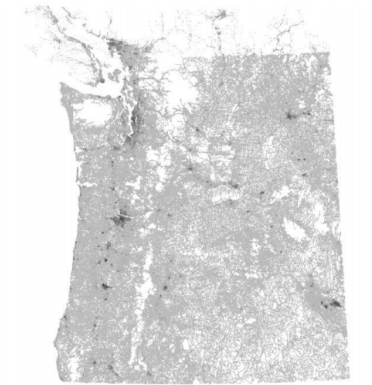
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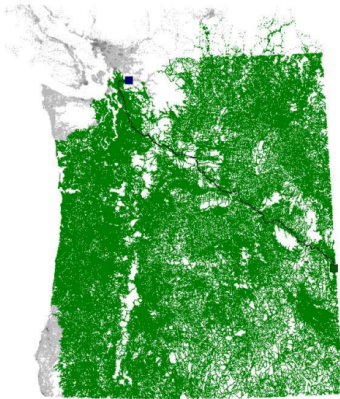
Road networks



1.6M vertices, 3.8M arcs, travel time metric.

Picture by Andrew Goldberg.

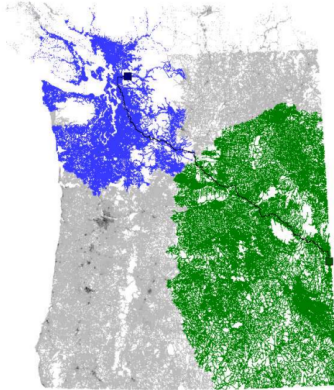
Road networks



Searched area

Picture by Andrew Goldberg.

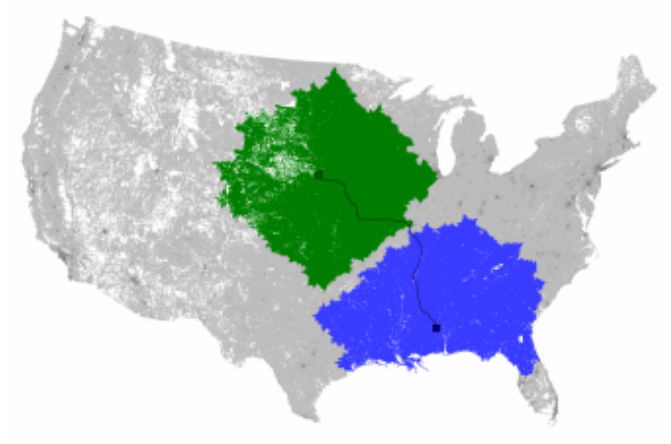
Road networks



forward search / reverse search

Picture by Andrew Goldberg.

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- Let's look at social networks

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- Can pass a message from any person to any person in at most 6 handshakes
- This is close to truth according to experiments and is called a “six handshakes” or “six degrees of separation” idea

Facebook

- Suppose an average person has around 100 Facebook friends

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Facebook

- Suppose an average person has around 100 Facebook friends
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- 1 trillion people at six handshakes
- Not possible, as there are only about 7 billion people on earth

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- Roughly 1M friends of friends of friends
- $1M + 1M = 2M$ people — 1000 times less

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- Instead of searching for all possible objects, search for first halves and for second halves separately
- Then find “compatible” halves
- Typically roughly $O(\sqrt{N})$ instead of $O(N)$, including the previous Facebook example

Conclusion

- Dijkstra goes in “circles”
- Bidirectional search idea can reduce the search space
- Roughly 2x speedup for road networks
- Meet-in-the-middle — $\sqrt{N}$ instead of N
- 1000 times faster for social networks
- Next video — Bidirectional Dijkstra algorithm

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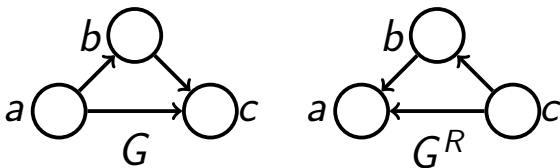
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- Repeat until t is processed

Reversed Graph

Definition

Reversed graph G^R for a graph G is the graph with the same set of vertices V and the set of reversed edges E^R , such that for any edge $(u, v) \in E$ there is an edge $(v, u) \in E^R$ and vice versa.



Bidirectional Dijkstra

- Build G^R

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Bidirectional Dijkstra

- Build G^R
- Start Dijkstra from s in G and from t in G^R
- Alternate between Dijkstra steps in G and in G^R
- Stop when some vertex v is processed both in G and in G^R
- Compute the shortest path between s and t

Computing Distance

Let v be the first vertex which is processed both in G and in G^R . Does it follow that there is a shortest path from s to t going through v ?

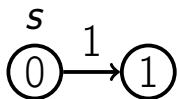
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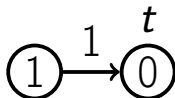
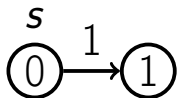
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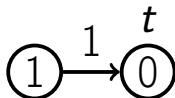
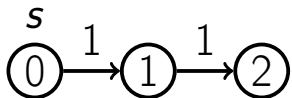
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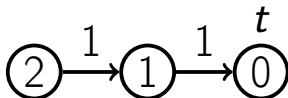
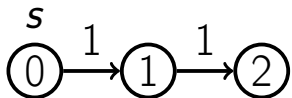
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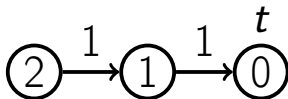
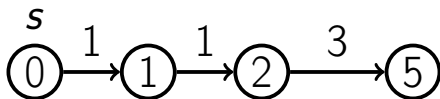
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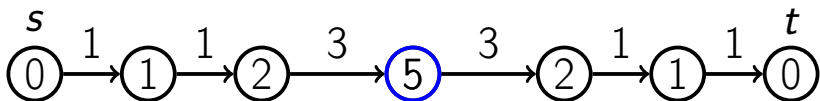
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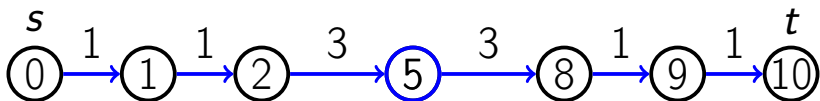
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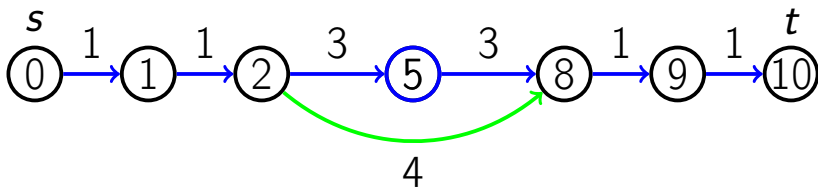
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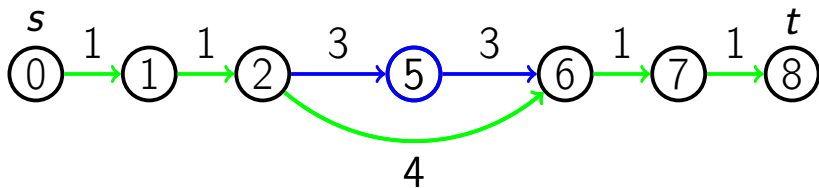
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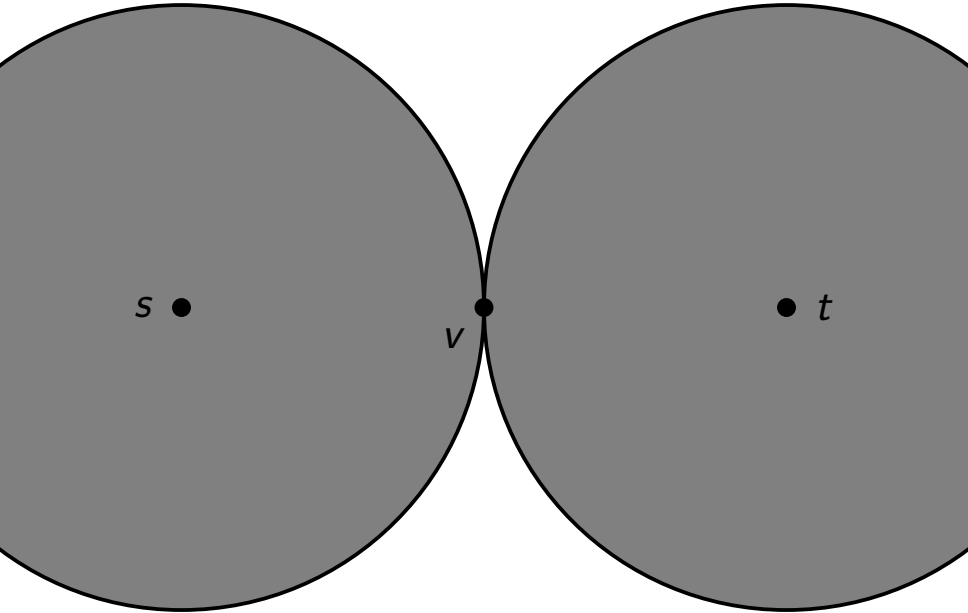


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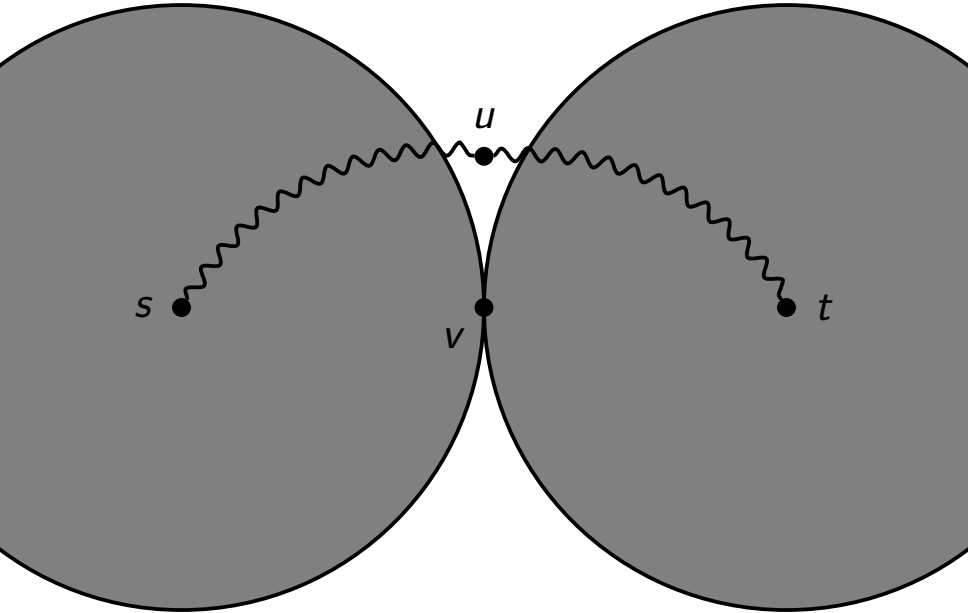
Lemma

Let $\text{dist}[u]$ be the distance estimate in the forward Dijkstra from s in G and $\text{dist}^R[u]$ — the same in the backward Dijkstra from t in G^R . After some node v is processed both in G and G^R , some shortest path from s to t passes through some node u which is processed either in G , in G^R , or both, and $d(s, t) = \text{dist}[u] + \text{dist}^R[u]$.

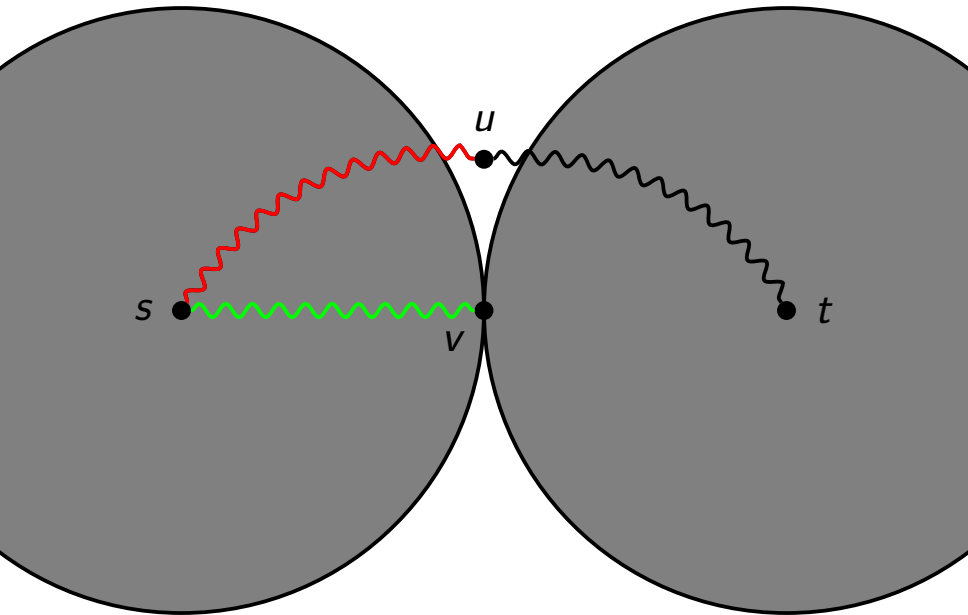
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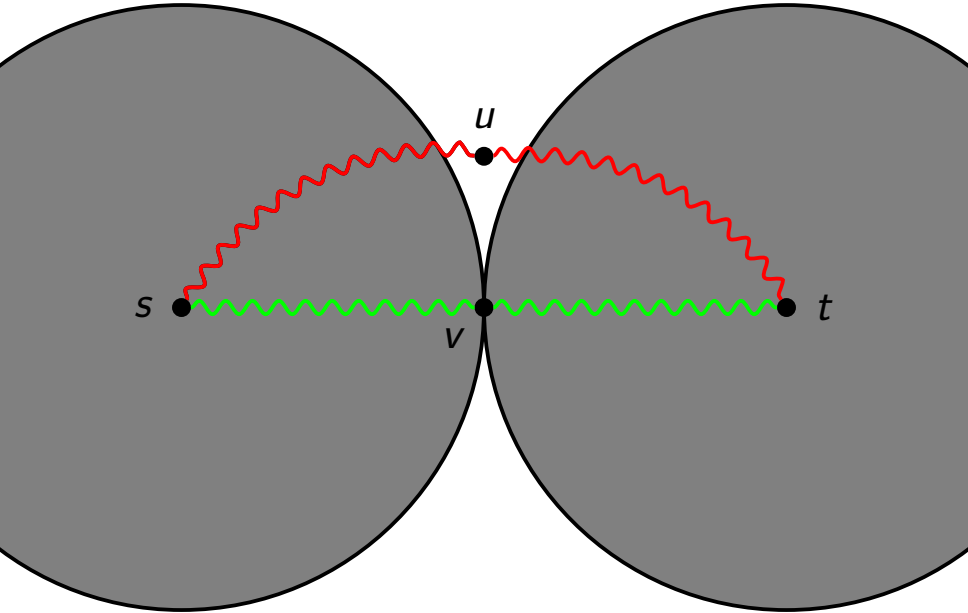
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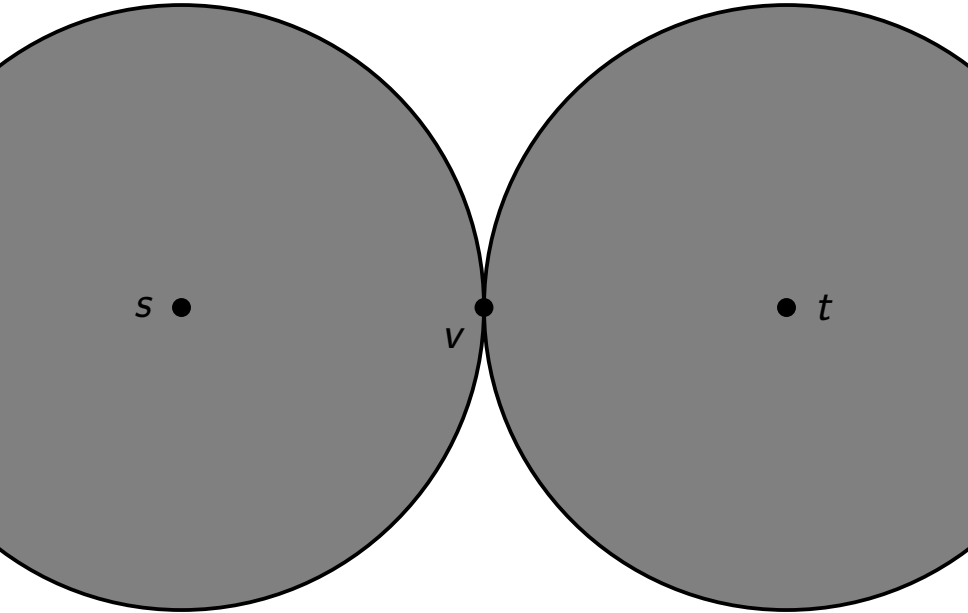
Proof



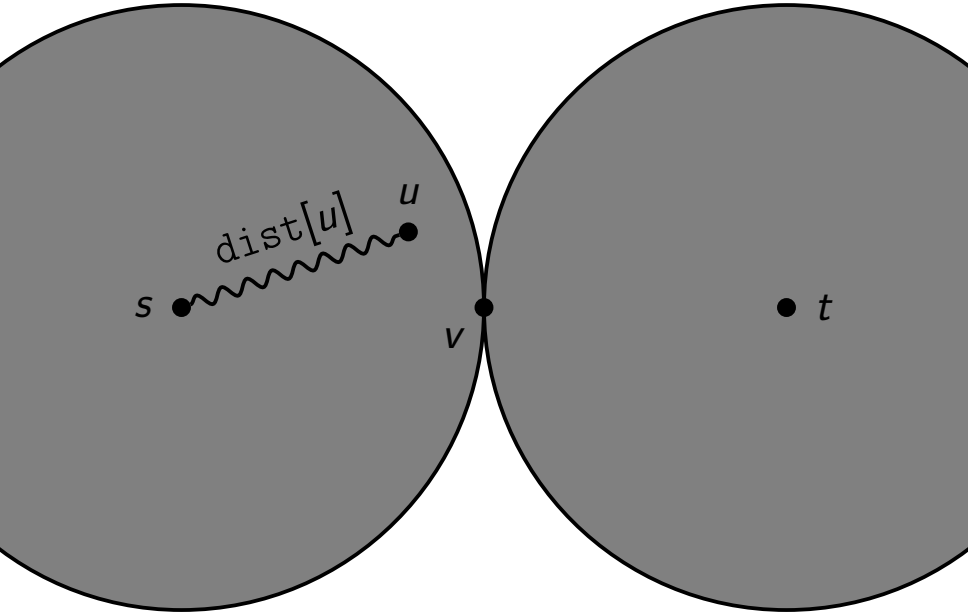
Proof



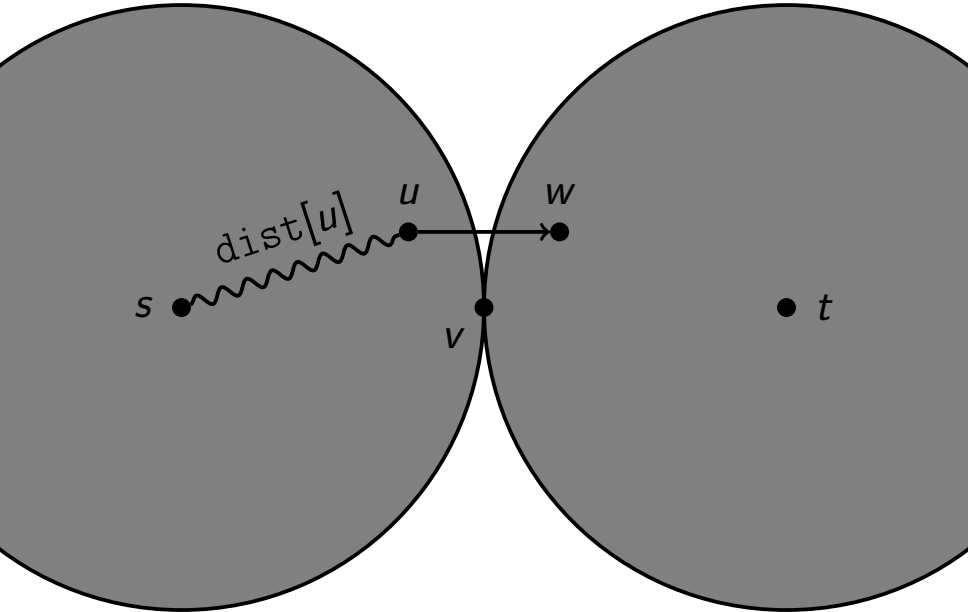
Proof



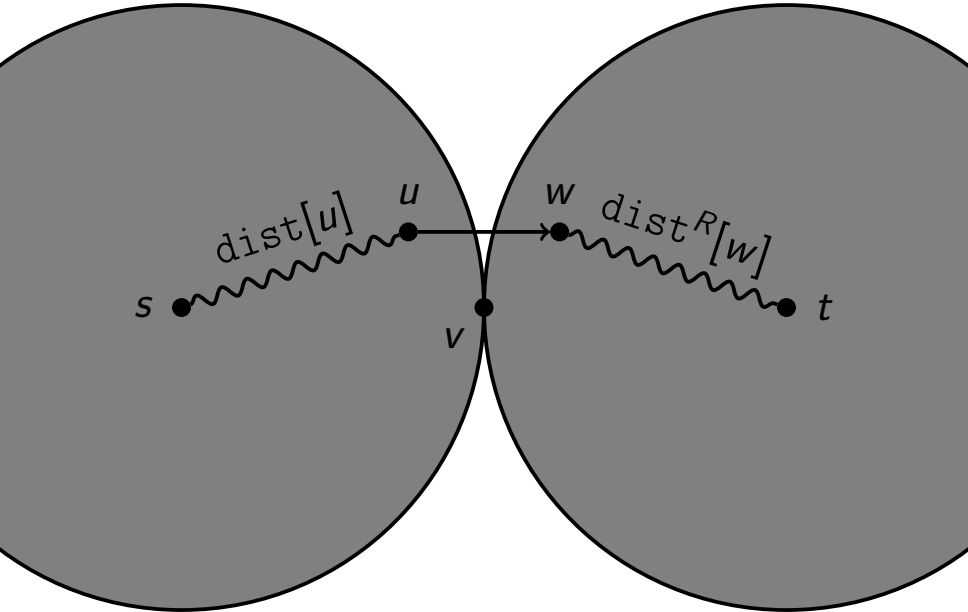
Proof



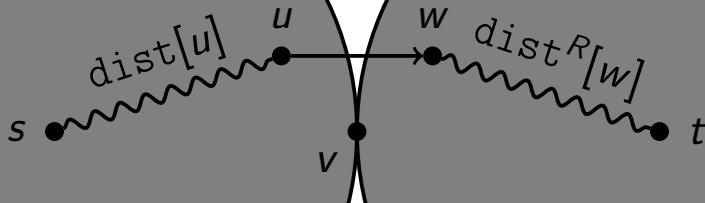
Proof



Proof

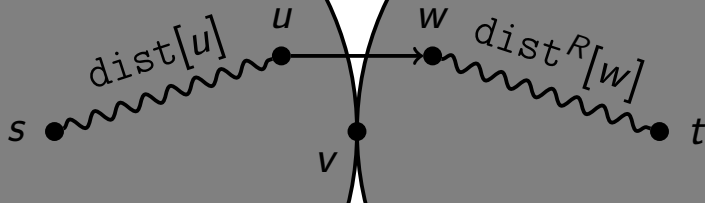


Proof



$$d(s, t) = \text{dist}[u] + l(u, w) + \text{dist}^R[w]$$

Proof



$$d(s, t) = \text{dist}[u] + l(u, w) + \text{dist}^R[w] =$$

$$= \text{dist}[u] + \text{dist}^R[u]$$

BidirectionalDijkstra(G, s, t)

```
 $G^R \leftarrow \text{ReverseGraph}(G)$   
Fill  $\text{dist}, \text{dist}^R$  with  $+\infty$  for each node  
 $\text{dist}[s] \leftarrow 0, \text{dist}^R[t] \leftarrow 0$   
Fill  $\text{prev}, \text{prev}^R$  with None for each node  
 $\text{proc} \leftarrow \text{empty}, \text{proc}^R \leftarrow \text{empty}$   
do:  
     $v \leftarrow \text{ExtractMin}(\text{dist})$   
     $\text{Process}(v, G, \text{dist}, \text{prev}, \text{proc})$   
    if  $v$  in  $\text{proc}^R$ :  
        return  $\text{ShortestPath}(s, \text{dist}, \text{prev}, \text{proc}, t, \dots)$   
     $v^R \leftarrow \text{ExtractMin}(\text{dist}^R)$   
    repeat symmetrically for  $v^R$  as for  $v$   
while True
```

Relax($u, v, \text{dist}, \text{prev}$)

```
if  $\text{dist}[v] > \text{dist}[u] + w(u, v)$ :  
     $\text{dist}[v] \leftarrow \text{dist}[u] + w(u, v)$   
     $\text{prev}[v] \leftarrow u$ 
```

Process(u , G , dist, prev, proc)

for $(u, v) \in E(G)$:

 Relax(u , v , dist, prev)

proc.Append(u)

ShortestPath($s, \text{dist}, \text{prev}, \text{proc}, t, \text{dist}^R, \text{prev}^R, \text{proc}^R$)

```
 $distance \leftarrow +\infty, u_{best} \leftarrow \text{None}$   
for  $u$  in  $\text{proc} + \text{proc}^R$ :  
    if  $\text{dist}[u] + \text{dist}^R[u] < distance$ :  
         $u_{best} \leftarrow u$   
         $distance \leftarrow \text{dist}[u] + \text{dist}^R[u]$   
 $path \leftarrow \text{empty}$   
 $last \leftarrow u_{best}$   
while  $last \neq s$ :  
     $path.\text{Append}(last)$   
     $last \leftarrow \text{prev}[last]$   
 $path \leftarrow \text{Reverse}(path)$   
 $last \leftarrow u_{best}$   
while  $last \neq t$ :  
     $last \leftarrow \text{prev}^R[last]$   
     $path.\text{Append}(last)$   
return ( $distance, path$ )
```

Conclusion

- Worst-case running time of Bidirectional Dijkstra is the same as for Dijkstra
- Speedup in practice depends on the graph
- Memory consumption is 2x to store G and G^R
- You'll see the speedup on social network graph in the Programming Assignment